p-adic Analysis 5

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Problem 2

Let $x = 9/16$ and $y = 9/4$. The p-adic logarithm is

$$\log(1+z) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{z^n}{n},$$

so

$$\sum_{n=1}^{\infty} (-1)^n \frac{3^{2n}}{n4^{2n}} = \sum_{n=1}^{\infty} (-1)^n \frac{x^n}{n} = -\log(1+x).$$

Since $x = 9/16$,

$$1 + x = \frac{25}{16} = \left(\frac{5}{4}\right)^2,$$

$$-\log(1+x) = -\log\left(\left(\frac{5}{4}\right)^2\right).$$

Also,

$$1 - y = -\frac{5}{4},$$

so

$$(1-y)^2 = \frac{25}{16} = 1 + x.$$

Therefore,

$$-\log(1+x) = -\log((1-y)^2) = -2\log(1-y)$$

$$= -2\sum_{n=1}^{\infty} \frac{3^{2n}}{n4^n}.$$

This highlights that $\log(-5/4) = \log(5/4)$ because $\log(-1) = 0$.

Problem 4

For $n \geq 2$,

$$v_p\left(\frac{p^n}{n!}\right) \geq n - v_p(n!) \geq 2,$$

so

$$e^p \equiv 1 + p \pmod{p^2}.$$

For contradiction, assume that $x^p \equiv e^p \pmod{p^2}$ for some $x \in \mathbb{Z}_p$. Since

$$e^p \equiv 1 + p \equiv 1 \pmod{p},$$

we have that

$$x^p \equiv 1 \pmod{p}.$$

Also,

$$x^p \equiv x \pmod{p},$$
$$x \equiv 1 \pmod{p}.$$

So we can write $x = 1 + pa$ with $a \in \mathbb{Z}_p$. So

$$x^p = 1 + p(pa) + \binom{p}{2}p^2a^2 + \cdots \equiv 1 \pmod{p^2}.$$

But we already found $e^p \equiv x^p \equiv 1 + p \pmod{p^2}$, which is a contradiction.

Problem 5

Assume $x \equiv y \pmod{p^N}$, so that x and y have the same first N digits. That means $f(x)$ and $f(y)$ have the same digit contributions in positions

$$0^2, 1^2, \ldots, (N-1)^2,$$

so

$$f(x) \equiv f(y) \pmod{p^{N^2}}.$$

Therefore,

$$|x - y|_p \leq p^{-N} \implies |f(x) - f(y)|_p \leq p^{-N^2},$$

which proves continuity.

For differentiability, let $h \in p^N\mathbb{Z}_p$. Then x and $x + h$ have the same first N digits. Therefore,

$$f(x + h) - f(x) \in p^{N^2}\mathbb{Z}_p,$$

so

$$\left|\frac{f(x+h) - f(x)}{h}\right|_p \leq p^{-N^2+N} \to 0,$$

so $f'(x) = 0$.

Problem 12

For any $x \in B$ and $m \geq 0$, we have that

$$f(x + mk) = f(x).$$

Now, fix $x \in B$. For all $r \geq 0$, we can take $m = p^r$, so that

$$f(x + p^r k) = f(x),$$

$$\frac{f(x + p^r k) - f(x)}{p^r k} = 0.$$

As $r \to \infty$, $|p^r k|_p \to 0$, so $p^r k \to 0$. So the derivative of f at x can be written as

$$f'(x) = \lim_{r \to \infty} = \lim_{r \to \infty} \frac{f(x + p^r k) - f(x)}{p^r k} = 0.$$

So $f'(x) = 0$ for all $x \in B$. Since f can be written as a power series, all coefficients of terms of degree ≥ 1 must be 0, and f is constant.

Problem 14

Take $B = \mathbb{Z}_p$ and $f(x) = x^2$. This f is analytic on B. Now, choose $a = 0$, $b = 1$, $n = -1$. Then

$$|f(a)|_p = 0, \quad |f(b)|_p = |1|_p = 1.$$

So it is true that

$$|f(a)|_p \leq p^{-1} \leq |f(b)|_b.$$

IVT in p-adics would imply that there exists $c \in \mathbb{Z}_p$ where

$$|f(c)|_p = p^{-1},$$

but

$$|f(c)|_p = |c^2|_p = |c|_p^2 \in \{0, 1, p^{-2}, p^{-4}, \ldots\},$$

where none of the options are p^{-1}. So IVT fails, even for analytic functions.

Problem 16

Take $B = \mathbb{Z}_p$ and $f(x) = x^p$, which is analytic. Choose $a = 0$, $b = 1$. So

$$\frac{f(b) - f(a)}{b - a} = 1.$$

Assuming MVT in p-adics would imply that there is $x \in \mathbb{Z}_p$ such that $f'(x) = 1$. But $f'(x) = px^{p-1}$, so

$$x^{p-1} = \frac{1}{p}.$$

Since $|x|_p \leq 1$,

$$|x^{p-1}|_p = |x|_p^{p-1} \leq 1,$$

but

$$\left|\frac{1}{p}\right|_p = p > 1.$$

This is a contradiction, so no such x exists and MVT is not true for p-adics.